Project Reflection – A03: Functional Code Implementation

Mode Share Dashboard (City of Vancouver)

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1. Work Process

How did the transition from design to code change your approach to problem-solving? Consider specific moments where you had to adapt to accommodate technical constraints or where coding revealed new design possibilities.

The transition from design to code changed how I solved problems because I had to think about making things actually work instead of just how they look. The biggest challenge was connecting the Leaflet map with the indicator-data.json file. I knew what I wanted it to do from my design, but making the polygons clickable and linking them to the chart data was confusing at first. I used AI to help understand how to structure the code, and after studying the examples, I figured out how to implement it myself. Another challenge was the accordion sections. In Figma, they looked simple, but coding the toggle functionality required understanding event listeners and how to change the aria-expanded attributes for accessibility. I also had a CSS issue where the charts kept growing in height automatically and wouldn't stop. It took me a while to realize I had set auto height, and once I fixed that with proper height constraints, everything worked properly.

To what extent did design approach continue to play a role during the coding phase? Reflect on whether you approached development as pure implementation or if you continued to iterate and refine design decisions while coding.

For this project, I approached coding mostly as implementation of what I already designed in Figma. I made all my design decisions during the Hi-Fi phase, so during coding I focused on making those designs work functionally. My goal was to match the visual design as closely as possible with the code. I didn't make major design changes while coding because I wanted to stay true to what Char and I had already planned and finalized together. The styling in CSS was about recreating what I had in Figma, like the colors, spacing, typography, and button styles. However, there were small technical adjustments I had to make, like setting proper container heights for the charts and adjusting some padding to make sure the accordion animations looked smooth. But these were more about making the design work properly in the browser rather than changing the actual design itself.

How did working with data and functional code influence your understanding of the design decisions you made earlier?

Working with real JSON data helped me understand the importance of designing with actual content in mind. When I connected data.json to the mode share bar chart, I could see how the actual numbers looked compared to my placeholder data in Figma. The indicator-data.json file was more complex because it had geographical data with coordinates and multiple records for different areas. Implementing this made me realize that data structure really matters when designing dashboards. I also learned that having two separate JSON files meant I needed to fetch and process them differently, which added complexity I hadn't fully considered during design. Seeing the charts render with real data also showed me that my design choices for chart colors and sizing worked well. The experience taught me that in future projects, I should understand the data structure better during the design phase so I can anticipate technical challenges earlier.

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2. Individual Role in the Group Project

How did your contributions to the project shift between the design phase (up to A02) and the integration phase? Be specific about whether you took on new responsibilities, deepened existing ones, or moved between design and development tasks.

My role shifted from being the design lead in A02 to being the coding lead in A03. During the design phase, I focused on creating wireframes, doing the Crazy 8 exercises with Char, and building the Hi-Fi prototype in Figma. I handled most of the visual design work while Char managed documentation and research. For A03, I took responsibility for writing the code to make our design functional. I wrote the JavaScript in index.js, including the accordion toggle behavior, the data fetching from both JSON files, and connecting the Leaflet map with the chart updates. I also handled the CSS styling to match our Figma design. The most challenging parts were implementing the interactive map with clickable polygons and getting both Chart.js visualizations to work with the data. While I mentioned in my previous reflection that I have some development experience, this project really pushed me to learn new libraries like Leaflet and Chart.js that I hadn't used before.

Which phase of the project (design-focused or integration-focused) felt more natural to you, and why? Reflect on your comfort level, engagement, and the type of thinking required in each phase.

The design phase felt more natural to me because visual design is where I'm most comfortable. Working in Figma, creating layouts, choosing colors, and building prototypes matches how I think creatively. Like I mentioned in my previous reflection, I'm stronger on the design side. The coding phase was more challenging because it required different thinking. Instead of visual problem solving, I had to think logically about how functions work, how data flows through the code, and how to debug errors. There were frustrating moments, like when I couldn't figure out why the chart heights kept increasing or when the map data wasn't loading correctly. But there were also satisfying moments when things finally worked, like when I got the accordion animations smooth or when the map polygons correctly updated the chart. Even though coding pushed me out of my comfort zone, I learned a lot and felt accomplished when I solved problems, especially with help from AI tools to understand concepts I was confused about.

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3. Roles of Partners in the Group Project

How did the division of labor between design lead and technology lead evolve during the integration phase? Reflect on whether these roles remained distinct, blurred, or shifted entirely, and whether this evolution was productive.

The division of work stayed pretty clear during the integration phase, similar to how we worked in the design phase. I took the lead on coding and implementing all the interactive features, while Char continued her strengths in organization and documentation. She managed the GitHub repository, wrote the detailed README.md file with instructions on how to run the project locally, and handled deploying the site to Netlify. Char also did testing and would tell me when something wasn't working right or when the design didn't match our Figma mockup. Our communication through Discord stayed strong like I mentioned before, with regular updates about what we were working on and any issues we found. Char kept track of deadlines and made sure all our files were organized properly. This division worked well because we each focused on what we're good at. I could concentrate on solving the technical coding challenges while Char made sure everything was documented and submitted correctly. Working this way was productive and let us finish the project efficiently.